

$$p(\mu | x) = p(x | \mu) p(\mu) / K_1$$

$$= \exp \left[\frac{-n}{2\sigma^2} (\bar{x} - \mu)^2 - \frac{(\mu - \mu_0)^2}{2\sigma_0^2} \right]$$

$$= \exp \left[\frac{-n}{2\sigma^2} (\bar{x}^2 + \mu^2 - 2\bar{x}\mu) - \frac{(\mu^2 + \mu_0^2 - 2\mu\mu_0)}{2\sigma_0^2} \right]$$

$$= \exp \left[\frac{-n}{2\sigma^2} (\mu^2 - 2\bar{x}\mu) - \cancel{\frac{n\bar{x}^2}{2\sigma^2}} - \left(\frac{\mu^2 - 2\mu\mu_0}{2\sigma_0^2} \right) - \cancel{\frac{\mu_0^2}{2\sigma_0^2}} \right] K_2$$

$$= \exp \left[-\mu^2 \left(\frac{n}{2\sigma^2} + \frac{1}{2\sigma_0^2} \right) + 2\mu \left(\frac{n\bar{x}}{2\sigma^2} + \frac{\mu_0}{2\sigma_0^2} \right) \right] K_2$$

$$= \exp \left[- \left\{ \underbrace{\mu^2 \left(\frac{n}{2\sigma^2} + \frac{1}{2\sigma_0^2} \right)}_a - 2\mu \underbrace{\left(\frac{n\bar{x}}{2\sigma^2} + \frac{\mu_0}{2\sigma_0^2} \right)}_b \right\} \right] K_2$$

$$a\mu^2 - 2b\mu = a \left(\mu^2 - \frac{2b}{a}\mu \right) + \frac{b^2}{a^2} - \frac{b^2}{a^2}$$

$$= a \left(\mu - b/a \right)^2 + \text{constant}$$

$$= \exp \left[-a \left(\mu - b/a \right)^2 \right] K_3$$